

Algorithm Explanation:

Implementation of Kalman filter

Pseudocode:

Offset = initial_GPS_altitude - initial_altimeter_altitude

Loop through all values of GPS and Altimeters

Set Q, R values for Kalman filter based on the distance estimate to ground [Based on GPS-offset] [Prediction Step]

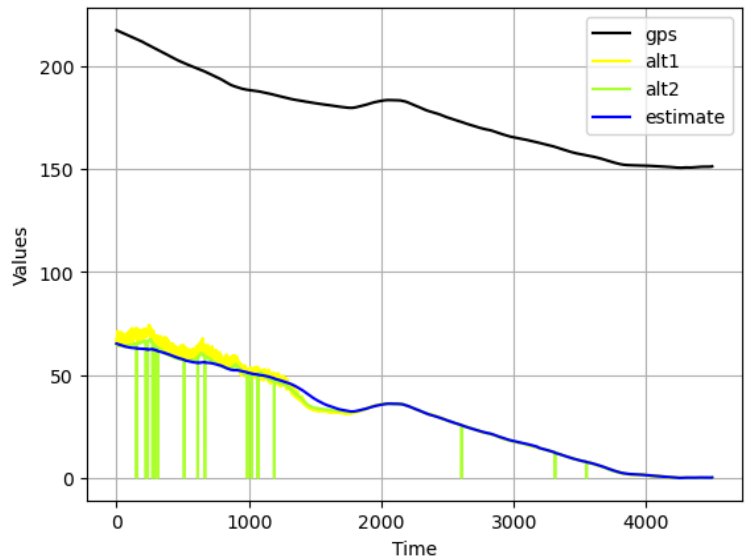
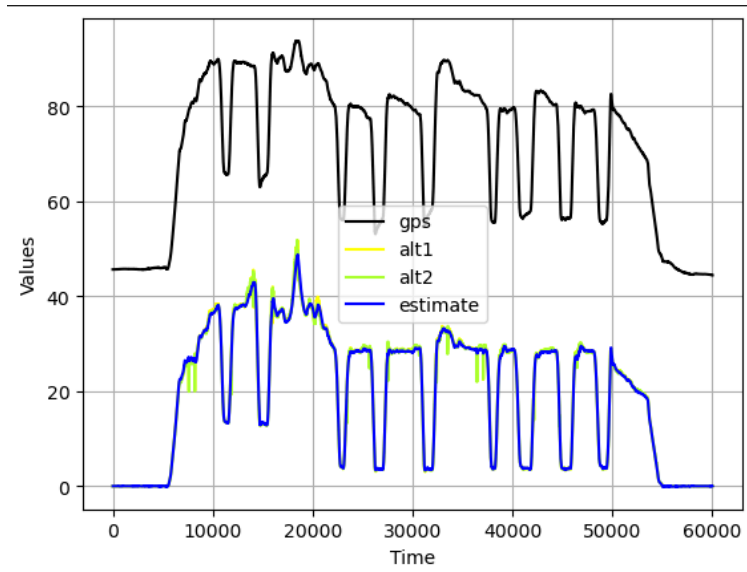
altitude_estimate = gps[i] - offset

Update altitude_estimate and covariance using altimeter 1 data

Update altitude_estimate and covariance using altimeter 2 data

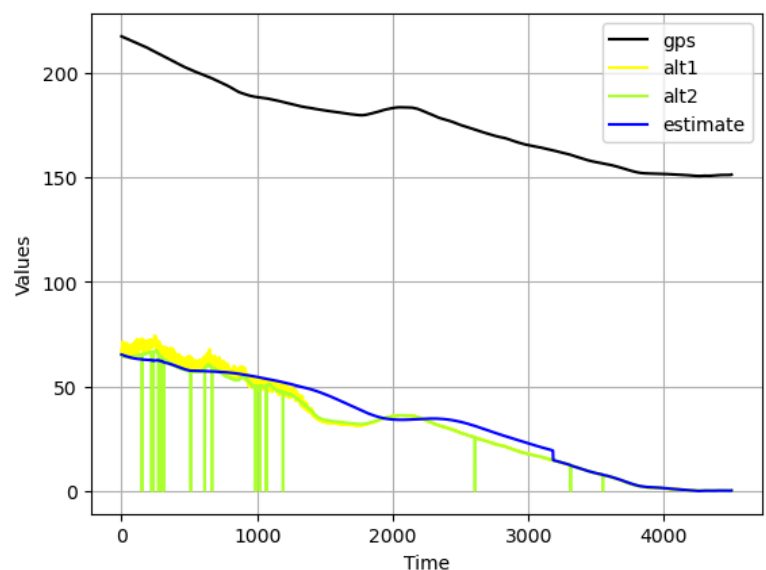
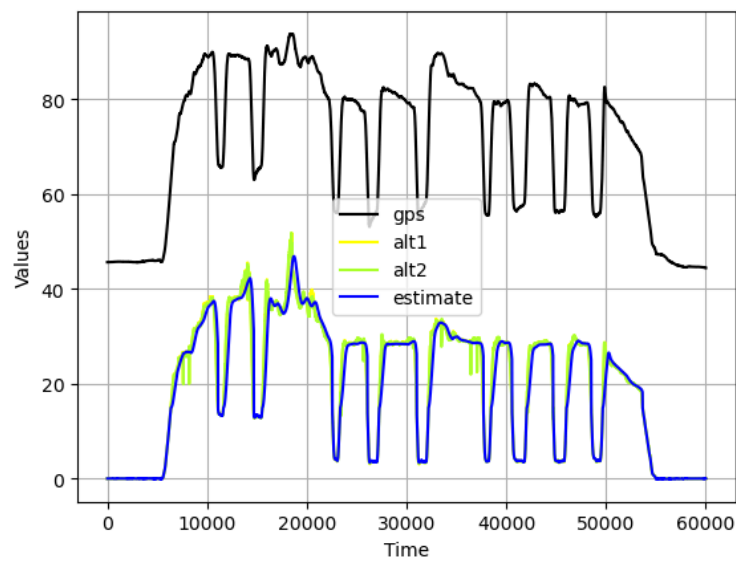
Recalculate offset

Kalman Filter:



With the Kalman filter, the altitude estimates are relatively smooth with the sudden spikes from the altimeter measurement removed. However, we do observe in Image 1 that there is scope to smoothen the estimate further.

Kalman filter with Moving Window Average for Smoothing:



To smoothen the estimate further, a moving window average filter is used alongside the Kalman filter. When a moving average with a window size of 250 is applied, although Image 1 shows a smoother trajectory we see in Image 2 that the moving window has brought a lag in the state estimation. We are seeing a discontinuity at about 3150 since I have provided the condition to change R, Q (noise) values for the Kalman filter if the estimate is less than 15 m since we need more accurate measurements below that altitude.

The final submitted code just has Kalman filter implementation and doesn't contain the moving average filter.